12. $\int udv = uv - \int vdu$

13. $\int udv = uv - u'v_1 + u''v_2 - u'''v_3 + \ldots$

14. $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + c$

15. $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c$

16. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \log \left| x + \sqrt{x^2 - a^2} \right| + c$

17. $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + c$

18. $\int e^x \left[f(x) + f'(x) \right] dx = e^x f(x) + c$

19. $\int e^{ax} \left[a f(x) + f'(x) \right] dx = e^{ax} f(x) + c$

20. $\int \sqrt{x^2 - a^2}\, dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + c$

21. $\int \sqrt{x^2 + a^2}\, dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log \left| x + \sqrt{x^2 + a^2} \right| + c$

- **Definite integral:**

Let $f(x)$ be a continuous function on [a,b] and if $F(x)$ is anti derivative of $f(x)$, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

- **Properties of definite integrals:**

(i) $\int_a^b f(x)dx = \int_a^b f(t)dt$ (ii) $\int_a^b f(x)dx = -\int_b^a f(x)dx$

(iii) $\int_a^b \left[f(x) \pm g(x) \right] dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$ (iv) $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

(v) $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ (vi) $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

(vii) a) If $f(x)$ is an even function, then $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$

b) If $f(x)$ is an odd function, then $\int_{-a}^{a} f(x)dx = 0$

- **Particular case of Gamma Integral:**

If n is a positive integer, then $\int_0^\infty x^n e^{-ax}\, dx = \frac{n!}{a^{n+1}}$

- **Properties of gamma function:**

(i) $\Gamma(n) = (n-1)\Gamma(n-1), n>1$ (ii) $\Gamma(n+1) = n\Gamma(n), n>0$

(iii) $\Gamma(n+1) = n!$, n is a positive integer (iv) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

- **Definite integral as the limit of a sum:**

Let $f(x)$ be a continuous real valued function in $[a, b]$, which is divided into n equal parts each of width h, then

$\int_a^b f(x)dx = \lim_{\substack{h\to 0 \\ n\to\infty}} \sum_{r=1}^{n} h\, f(a+rh)$ where $h = \frac{b-a}{n}$

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